# Eight qubits: exact continuation to 200,000 steps

Usual noise σ=0.05 | batch 65,536 | requested exponents 3/4, 1/4, 1/2

## The trajectory reaches 99% at saved step 174,000 and finishes at 99.13929%.

| Checkpoint | Fidelity | Sampled loss | TP residual |
|---|---|---|---|
| Parent: 148,000 | 98.76891% | 0.0011358295 | 0.1976805 |
| Continuation: 200,000 | 99.13929% | 0.00113493693 | 0.1706184 |

Answer: more iterations alone were sufficient for this requested-exponent trajectory to cross 99%. Every saved checkpoint from 191,000 through 200,000 remained above 99%, so the endpoint is not a lone saved-threshold excursion.

From 148,000 to 200,000 steps, fidelity improves by 0.3704 percentage points. The fixed-sample loss changes from 0.0011358295 to 0.00113493693; the complete-factor TP residual improves from 0.1976805 to 0.1706184. TP remains separate: fidelity is not a physical-feasibility certificate.

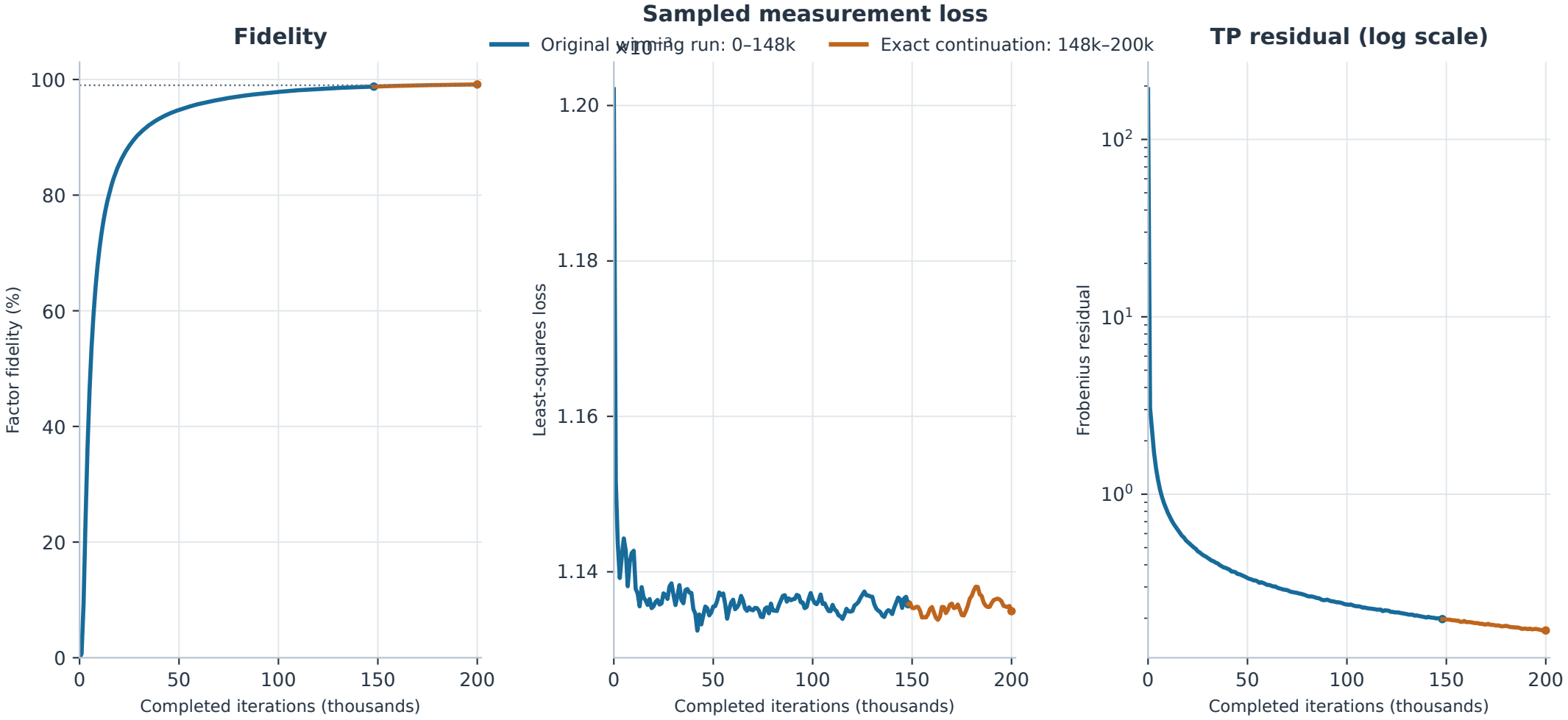
## Fixed scientific scope

One exact restart, not a new sweep: the parent factor, momentum estimate, PCG64 sampling state, global schedule index, dataset, frozen source snapshot and all optimizer settings are preserved. The segment adds 52,000 updates and 3,407,872,000 optimizer-row draws, ending at 13,107,200,000 total draws. No fidelity early stopping was used.

This result concerns one Haar-unitary channel, one noisy dataset and one seed. It does not establish a success probability, uncertainty interval or universal optimum.

# Combined saved trajectory

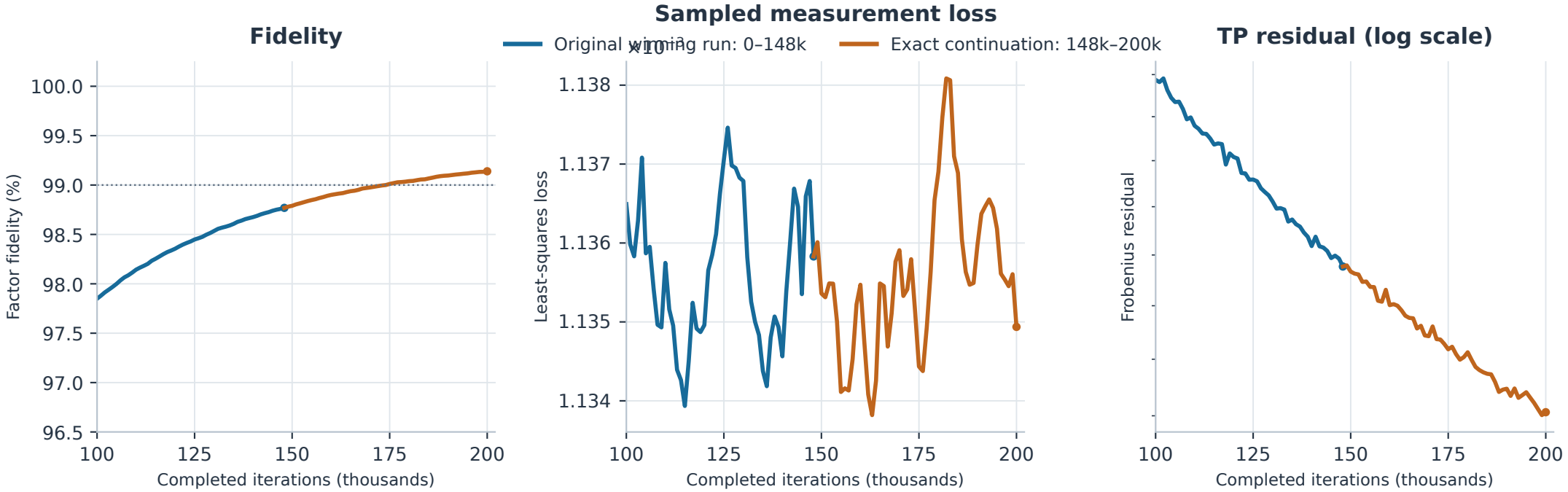
Solid colored trajectories; saved evidence at 1,000-step cadence; no smoothing or fitted curves




Fidelity is $|c^\dagger u|^2/(||c||^2||u||^2)$, computed from complete factors and invariant to nonzero phase/scale. Loss uses the fixed 512-row metric sample with seed 12345. TP is $||T(uu^\dagger)-I||_F$ from the complete factor and uses a logarithmic axis. The dotted line is the 99% reference.

# Late trajectory and threshold persistence

Zoomed to 100,000–200,000 steps; scientific traces remain unsmoothed




| Step | Fidelity | Sampled loss | TP residual |
|---|---|---|---|
| 191,000 | 99.10148% | 0.00113636912 | 0.1734496 |
| 192,000 | 99.10645% | 0.00113646555 | 0.1747628 |
| 193,000 | 99.11019% | 0.00113655213 | 0.1731092 |
| 194,000 | 99.11469% | 0.00113644414 | 0.1735892 |
| 195,000 | 99.11820% | 0.00113618074 | 0.1740994 |
| 196,000 | 99.12464% | 0.00113561092 | 0.17316 |
| 197,000 | 99.12849% | 0.00113553702 | 0.1722434 |
| 198,000 | 99.13288% | 0.00113545259 | 0.1711749 |
| 199,000 | 99.13407% | 0.00113560469 | 0.170114 |
| 200,000 | 99.13929% | 0.00113493693 | 0.1706184 |

## Observed threshold behavior

First saved F ≥ 99%: 174,000 steps. All saved checkpoints 191,000–200,000 ≥ 99%: yes. The prior descriptive projection suggested 170–175k; the observed saved crossing is 174k. This does not make the earlier fit a confidence interval. TP improves from 0.1976805 at restart to 0.1706184 at the endpoint.

# Exact schedules, budgets, and runtime definitions

Global schedule index is preserved across the restart

| Quantity | Exact value | Meaning |
|---|---|---|
| Step $\gamma_k$ | min(1, 0.5/(k+10)^(3/4)) | Frank-Wolfe atom weight |
| Smoothing $\beta_k$ | $10^7$/(k+1)^(1/4); no cap | TP penalty denominator |
| Momentum $\rho_k$ | min(1, 0.6/(k+4)^(1/2)) | New batch-gradient weight |
| Steps | 148,000 → 200,000 | 52,000 additional updates |
| Batch / rank / τ | 65,536 / 1 / 20 | 64-bit product-state backend |
| Metric | 512 rows every 1,000 steps | Fixed seed 12345 |

| Timer | Measured | Definition |
|---|---|---|
| Slurm elapsed | 12 min 17 s | Whole allocation; COMPLETED, exit 0:0 |
| Runner | 705.106 s (11.75 min) | After setup; includes compilation, preparation, transfers, metrics, restart writes |
| Optimizer | 403.873 s (6.73 min) | GPU execution for 52,000 updates |
| Host preparation | 346.519 s | Background time; overlaps GPU work |
| Metric evaluation | 2.358 s | Saved-checkpoint metrics |

Phase timers overlap and must not be added. Virtual $M=24^8=110,075,314,176$ rows is sampled with replacement and never materialized. The expected noisy-truth loss $\sigma^2/2=0.00125$ is not a measured convergence floor.

# Independent verification, provenance, and limitations

Complete saved factors; frozen source_fast; byte-for-byte Elite backup

| Check | Result |
|---|---|
| Scheduler | COMPLETED on ruche-gpu15; exit 0:0 |
| Restart continuity | Resumed factor hash and initial fidelity/loss/TP match parent endpoint |
| Exact schedules | All 52,000 applied scalar values match scales, offsets, exponents and caps |
| Independent factor metrics | Verified; primary ≥99% target met |
| Persistence | All saved checkpoints 191,000–200,000 ≥99% |
| Elite backup | All 11 copied files matched Ruche SHA-256 at collection |

## Independent numerical agreement

Saved fidelity 0.9913929063754052; NumPy factor form 0.9913929063754066; independent matrix form 0.9913929063754067. Saved TP 0.1706183674269944; independently computed TP 0.1706183674269938; truth TP residual 4.91e-14.

## Integrity note

The wrapper’s internal manifest matches every scientific artifact. Its Slurm stdout entry records the empty-file hash because checksums were computed before Slurm appended the final COMPLETE line. The finalized stdout was separately matched between Ruche and Elite; preserved evidence was not rewritten.

## Scope and limitations

One channel, noisy dataset, initialization and optimizer sampling seed. The result answers the bounded continuation question for this trajectory only. Fidelity is normalized overlap, not f in f+g and not a TP certificate; f is measurement least squares and g is the TP-constraint indicator.

**Frozen source** `/gpfs/workdir/silvetian/frankwolfe/results/qpt/n8_scale_tuning_20260923_01/source_fast`

**Final SHA-256** `6521b319b0e1cc68bb058c8be5e653b8aeac6bba35d724a2b588674681e89987`

**Resume factor** `fe2d5c002927690e6cdbc3501d0d38b73a1f628ecefc2dcf89ce96c07135589e`